

✓ MATH 152 Lab 2

Put team members' names and section number here.

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```
import sympy as sp
from sympy.plotting import (plot, plot_parametric)
```

Instructions: Complete the lab assignment in your assigned groups. Unless stated otherwise, your answers should be obtained using Python code.

Do not modify the cell above, as it contains all the packages you will need. It is highly recommended to not use any additional packages.

✓ Question 1

✓ 1a

```
x = sp.symbols('x')
radius = x
height = x - x**2
volume = 2 * sp.pi * sp.integrate(radius * height, (x, 0, 1))

print("exact answer:", volume)
print("as a decimal:", float(volume))
```

```
exact answer: pi/6
as a decimal: 0.5235987755982989
```

✓ 1b

```
x = sp.symbols('x')

height_left = x - 2
height_right = 4 - x

radius = x - 1

volume = 2*sp.pi*sp.integrate(radius*height_left, (x, 2, 3)) \
        + 2*sp.pi*sp.integrate(radius*height_right, (x, 3, 4))

print("exact answer:", volume)
print("as a decimal:", float(volume))
```

```
exact answer: 4*pi
as a decimal: 12.566370614359172
```

✓ 1c

```
x = sp.symbols('x')

radius = x
height = 2*sp.sqrt(1 - (x - 2)**2)

volume = (sp.pi/2) * sp.integrate(radius*height, (x, 1, 3))

print("exact answer:", sp.simplify(volume))
print("as a decimal:", float(volume))
```

```
exact answer: pi**2  
as a decimal: 9.869604401089358
```

✓ Question 2

✓ 2a

```
# Enter your code here  
x, p, q = sp.symbols('x p q')  
  
work = sp.integrate(p*x**2, (x,0,1)) \  
      + sp.integrate(q/sp.sqrt(x), (x,1,2))  
  
print("exact answer:", sp.simplify(work))
```

```
exact answer: p/3 - 2*q + 2*sqrt(2)*q
```

✓ 2b

```
# Enter your code here  
import sympy as sp  
x, w, m, l, d = sp.symbols('x w m l d')  
  
d = 10  
w = 25  
m = 15  
l = 10  
  
lin_m = m / l  
  
f_work = sp.integrate(lin_m * (1 - x) + w, (x, 0, d))  
  
print(f"{sp.simplify(f_work):.1f} ft-lb")
```

```
325.0 ft-lb
```

✓ 2c

```
# Enter your code here  
y = sp.symbols('y')  
  
vol = sp.integrate((sp.pi*(25-(y)**2))*(6-y), (y, 0, 5))  
  
tot_work = 1000*9.8*(vol)  
  
print(f"{tot_work:.1f} Joules")
```

```
10583240.3 Joules
```

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